

Trinity study list — score 5 and 4 (production only)

Tests, tools, and migrations are excluded.

Score is review effort from length + cyclomatic complexity + nesting — not a defect count.

- **44** functions at score **5** (Very high)
- **56** functions at score **4** (High)
- For each: what it does, why it is hard to study, and metrics.

Score 5 — Very high

1. `_Policy.scalar`

- **Score:** 5 · **CC:** 68 · **Nesting:** 4 · **LOC:** 110
- **Theme:** SQL / query runtime
- **Where:** [backend/src/trinity/queries/runtime/sql_policy.py:164](#)
- **What it does:** Walks one SQL expression tree and decides if each node is an allowed scalar (columns, literals, ops, aggregates).
- **Why it is hard:** Huge branch table over SQLGlot node types; one wrong case changes what user SQL can do. (CC=68 (many branches), 110 lines)

2. `recorded`

- **Score:** 5 · **CC:** 61 · **Nesting:** 2 · **LOC:** 93
- **Theme:** Publication / refresh
- **Where:** [backend/src/trinity/publication/checks.py:12](#)
- **What it does:** Rebuilds required/diagnostic check outcomes from immutable DB rows; no remote I/O.
- **Why it is hard:** Must mirror connector rules exactly without a second registry; many incomplete/mixed-proof cases. (CC=61 (many branches), 93 lines)

3. `verify_validation`

- **Score:** 5 · **CC:** 60 · **Nesting:** 3 · **LOC:** 95
- **Theme:** EIA connector / validate
- **Where:** [backend/src/trinity/connector/validate.py:817](#)
- **What it does:** Rechecks a passing validation receipt and files before a later stage uses them.
- **Why it is hard:** Offline handoff guard: many ways data or evidence can change or go incomplete after success. (CC=60 (many branches), 95 lines)

4. `_Policy.normalize`

- **Score:** 5 · **CC:** 54 · **Nesting:** 5 · **LOC:** 95
- **Theme:** SQL / query runtime
- **Where:** [backend/src/trinity/queries/runtime/sql_policy.py:275](#)
- **What it does:** Turns a validated SELECT into the closed column/alias list Trinity stores and executes.
- **Why it is hard:** Many projection/alias edge cases; output must stay exact for later cursors and schema checks. (CC=54 (many branches), nesting=5, 95 lines)

5. `_source_complete`

- **Score:** 5 · **CC:** 50 · **Nesting:** 3 · **LOC:** 85
- **Theme:** EIA connector / validate
- **Where:** [backend/src/trinity/connector/validate.py:283](#)
- **What it does:** Decides if EIA retrieval finished (progress, stable totals, empty-page exhaustion).
- **Why it is hard:** Completion mixes attempt status, route, totals, and page exhaustion—easy to misread. (CC=50 (many branches), 85 lines)

6. `supervise`

- **Score:** 5 · **CC:** 44 · **Nesting:** 7 · **LOC:** 139
- **Theme:** EIA connector / validate
- **Where:** [backend/src/trinity/connector/prepare.py:247](#)
- **What it does:** Parents a prepare child with stage/overall deadlines and retained evidence.
- **Why it is hard:** Deep nested stage machine plus deadlines and fsync handoffs; wrong order breaks custody. (CC=44 (many branches), nesting=7, 139 lines)

7. `read_verified_diagnostics`

- **Score:** 5 · **CC:** 41 · **Nesting:** 2 · **LOC:** 115
- **Theme:** Publication / refresh
- **Where:** [backend/src/trinity/publication/diagnostics.py:41](#)
- **What it does:** Verifies bundle/member hashes and complete evaluations before keeping diagnostic notes.
- **Why it is hard:** Hash identity plus embedded summaries; must reject incomplete evaluations without remote repair. (CC=41 (many branches), 115 lines)

8. `NationalChart`

- **Score:** 5 · **CC:** 37 · **Nesting:** 3 · **LOC:** 142
- **Theme:** Frontend UI
- **Where:** [frontend/src/pages/dashboard/NationalChart.tsx:26](#)
- **What it does:** Renders the national outage SVG chart, selection, and responsive width mapping.
- **Why it is hard:** Long UI function mixing layout math, selection state, and series rendering. (CC=37 (many branches), 142 lines)

9. `DashboardData`

- **Score:** 5 · **CC:** 36 · **Nesting:** 2 · **LOC:** 54
- **Theme:** Frontend UI
- **Where:** [frontend/src/pages/dashboard/Dashboard.tsx:95](#)
- **What it does:** Composes dashboard chart/table view, day selection, and contribution panel wiring.
- **Why it is hard:** Many UI branches for role, view mode, selection, and updating states in one component. (CC=36 (many branches))

10. `_validate_page`

- **Score:** 5 · **CC:** 34 · **Nesting:** 4 · **LOC:** 79
- **Theme:** EIA connector / validate
- **Where:** [backend/src/trinity/connector/client.py:174](#)
- **What it does:** Validates one sanitized EIA JSON page into typed source rows and metadata.
- **Why it is hard:** Many shape/window/unit/key checks; API error vs client error boundaries matter. (CC=34 (many branches), 79 lines)

11. `_Policy.__init__`

- **Score:** 5 · **CC:** 34 · **Nesting:** 2 · **LOC:** 38
- **Theme:** SQL / query runtime
- **Where:** [backend/src/trinity/queries/runtime/sql_policy.py:113](#)
- **What it does:** Tokenizes SQL and rejects disallowed shapes before deeper policy walks.
- **Why it is hard:** Dense early rejects (tokens, keywords, nesting flags); easy to miss which gate failed. (CC=34 (many branches))

12. `_bound_results`

- **Score:** 5 · **CC:** 34 · **Nesting:** 3 · **LOC:** 34
- **Theme:** EIA connector / validate
- **Where:** [backend/src/trinity/connector/validate.py:142](#)
- **What it does:** Checks check-result list identity (count, types, version/attempt/manifest binding) before status.
- **Why it is hard:** Strict structural predicates; failure modes look similar until you compare every field. (CC=34 (many branches))

13. `SafeTransport.__call__`

- **Score:** 5 · **CC:** 33 · **Nesting:** 2 · **LOC:** 71
- **Theme:** Errors / transport
- **Where:** [backend/src/trinity/errors.py:72](#)
- **What it does:** ASGI middleware that wraps HTTP with request IDs and safe error responses.
- **Why it is hard:** Must handle send-already-started cases without leaking internals. (CC=33 (many branches), 71 lines)

14. `read_candidate`

- **Score:** 5 · **CC:** 33 · **Nesting:** 4 · **LOC:** 62
- **Theme:** Publication / refresh
- **Where:** [backend/src/trinity/refresh/candidates.py:114](#)
- **What it does:** Authorizes, then reads candidate evidence and actions in one DB snapshot.
- **Why it is hard:** Auth-first then multi-table snapshot; eligibility and evidence must stay consistent. (CC=33 (many branches))

15. `_diagnostic`

- **Score:** 5 · **CC:** 31 · **Nesting:** 10 · **LOC:** 80
- **Theme:** EIA connector / validate
- **Where:** [backend/src/trinity/connector/validate.py:458](#)
- **What it does:** Evaluates one diagnostic code/scope and keeps exact observations / not-applicable reasons.
- **Why it is hard:** Deep nesting across diagnostic codes; each code has different applicability rules. (CC=31 (many branches), nesting=10, 80 lines)

16. `Docker._verify_isolation`

- **Score:** 5 · **CC:** 31 · **Nesting:** 1 · **LOC:** 22
- **Theme:** Storage / adapters
- **Where:** [backend/src/trinity/adapters/docker.py:107](#)
- **What it does:** Checks real container image, labels, mounts, and resource limits for query isolation.
- **Why it is hard:** Short but dense: many host/config fields must match the reservation exactly. (CC=31 (many branches))

17. `verify_stored_candidate`

- **Score:** 5 · **CC:** 29 · **Nesting:** 1 · **LOC:** 79
- **Theme:** EIA connector / validate
- **Where:** [backend/src/trinity/connector/pipeline.py:358](#)
- **What it does:** Rechecks local validation receipt and every remote bundle object before trusting storage.
- **Why it is hard:** Joins local and S3 proofs; missing completion or changed bytes must fail closed. (CC=29, 79 lines)

18. `SqlExplorer`

- **Score:** 5 · **CC:** 29 · **Nesting:** 3 · **LOC:** 59
- **Theme:** Frontend UI
- **Where:** [frontend/src/pages/sql/SqlExplorer.tsx:39](#)
- **What it does:** SQL workspace UI: edit, run, wait, show results/errors with role gating.
- **Why it is hard:** Busy/wait/error/result state machine plus session permissions in one component. (CC=29)

19. `PreviewOperation.__post_init__`

- **Score:** 5 · **CC:** 28 · **Nesting:** 2 · **LOC:** 42
- **Theme:** Other backend
- **Where:** [backend/src/trinity/contracts/queries.py:77](#)
- **What it does:** Validates preview operation fields (dates, IDs, dataset, page size) at construction.
- **Why it is hard:** Many typed rejects packed into post-init; invalid combinations look alike. (CC=28)

20. `stage_query`

- **Score:** 5 · **CC:** 27 · **Nesting:** 2 · **LOC:** 70
- **Theme:** SQL / query runtime
- **Where:** [backend/src/trinity/queries/staging.py:98](#)
- **What it does:** Verifies pinned manifest and every selected file, then seals a staged query request.
- **Why it is hard:** Path/symlink/digest checks before isolation; one miss opens unpublished data risk. (CC=27, 70 lines)

21. `PublicationRecovery.reconcile`

- **Score:** 5 · **CC:** 27 · **Nesting:** 2 · **LOC:** 62
- **Theme:** Publication / refresh
- **Where:** [backend/src/trinity/publication/recovery.py:40](#)
- **What it does:** Idempotent publication failure recovery under lock; safe to replay lost responses.
- **Why it is hard:** Must not move the active pointer if already published; generation/fence cases stack up. (CC=27)

22. `ExecutionService.event`

- **Score:** 5 · **CC:** 27 · **Nesting:** 6 · **LOC:** 61
- **Theme:** Publication / refresh
- **Where:** [backend/src/trinity/refresh/execution.py:145](#)
- **What it does:** Commits stage/attempt custody for worker events before acknowledging the supervisor.
- **Why it is hard:** Event-kind switch with ownership/fence checks; wrong order breaks refresh custody. (CC=27, nesting=6)

23. `verified_chunks`

- **Score:** 5 · **CC:** 27 · **Nesting:** 3 · **LOC:** 59
- **Theme:** Storage / adapters
- **Where:** [backend/src/trinity/adapters/s3.py:134](#)
- **What it does:** Streams object bytes in bounded chunks and verifies identities before completion.
- **Why it is hard:** Encoding chosen by trusted bundle, not S3 metadata; chunk and total limits interact. (CC=27)

24. `read_preview_publication`

- **Score:** 5 · **CC:** 27 · **Nesting:** 2 · **LOC:** 49
- **Theme:** Publication / refresh
- **Where:** [backend/src/trinity/publication/repository.py:74](#)
- **What it does:** Requires complete frozen evidence authority for an already-pinned version.
- **Why it is hard:** Candidate receipt alone is not enough; snapshot coupling with pin read is subtle. (CC=27)

25. `DatasetTable`

- **Score:** 5 · **CC:** 26 · **Nesting:** 3 · **LOC:** 86
- **Theme:** Frontend UI
- **Where:** [frontend/src/pages/catalog/TableView.tsx:24](#)
- **What it does:** Catalog table: filters, paging, facility/generator choices, preview fetch.
- **Why it is hard:** URL search params, React Query cache, and filter state all meet here. (CC=26, 86 lines)

26. `_snapshot`

- **Score:** 5 · **CC:** 26 · **Nesting:** 3 · **LOC:** 51
- **Theme:** EIA connector / validate
- **Where:** [backend/src/trinity/connector/validate.py:540](#)
- **What it does:** Reads only declared parquet/source files and keeps per-dataset integrity failures visible.
- **Why it is hard:** Partial integrity across datasets; must not hide one dataset failure behind another. (CC=26)

27. `_verify_receipt`

- **Score:** 5 · **CC:** 26 · **Nesting:** 2 · **LOC:** 50
- **Theme:** EIA connector / validate
- **Where:** [backend/src/trinity/connector/prepare.py:195](#)
- **What it does:** After child exit, matches retained receipt to manifest/bundle/storage completion.
- **Why it is hard:** Parent must trust a small receipt only; many mismatch cases without redoing network I/O. (CC=26)

28. Manifest.__post_init__

- **Score:** 5 · **CC:** 26 · **Nesting:** 2 · **LOC:** 41
- **Theme:** Other backend
- **Where:** [backend/src/trinity/contracts/manifest.py:159](#)
- **What it does:** Validates manifest identity/format/contract fields at construction.
- **Why it is hard:** Many typed invariants; failures collapse to ManifestError codes. (CC=26)

29. build_choice_response

- **Score:** 5 · **CC:** 25 · **Nesting:** 2 · **LOC:** 38
- **Theme:** SQL / query runtime
- **Where:** [backend/src/trinity/queries/choice_schemas.py:51](#)
- **What it does:** Builds a signed choices page after binding/order/parent/search checks.
- **Why it is hard:** Must not dedupe or drop invalid rows (that can hide truncation bugs). (CC=25)

30. S3Settings.__post_init__

- **Score:** 5 · **CC:** 25 · **Nesting:** 2 · **LOC:** 31
- **Theme:** Other backend
- **Where:** [backend/src/trinity/config.py:76](#)
- **What it does:** Validates S3 bucket/prefix/endpoint settings at startup.
- **Why it is hard:** Strict regex and path-safety rules packed into post-init. (CC=25)

31. main

- **Score:** 5 · **CC:** 24 · **Nesting:** 3 · **LOC:** 47
- **Theme:** Auth / sessions
- **Where:** [backend/src/trinity/auth/check.py:154](#)
- **What it does:** Terminal persona checker: login and optional catalog/preview/schedule probes.
- **Why it is hard:** Role matrix of expected pass/fail; must print only safe summaries. (CC=24)

32. _refresh_stopped

- **Score:** 5 · **CC:** 24 · **Nesting:** 1 · **LOC:** 42
- **Theme:** Publication / refresh
- **Where:** [backend/src/trinity/refresh/recovery.py:19](#)
- **What it does:** Detects only failure shapes written by current Refresh writers.
- **Why it is hard:** Pre-claim vs started failure vs lease/deadline look similar but differ in proof. (CC=24)

33. recovery_actions

- **Score:** 5 · **CC:** 24 · **Nesting:** 2 · **LOC:** 38
- **Theme:** Publication / refresh
- **Where:** [backend/src/trinity/refresh/recovery.py:63](#)
- **What it does:** Computes rerun/delete flags for a retained failed lifecycle.
- **Why it is hard:** Needs matching warning and admission owner; must preserve publications. (CC=24)

34. PublicationService.finish

- **Score:** 5 · **CC:** 23 · **Nesting:** 1 · **LOC:** 54
- **Theme:** Publication / refresh
- **Where:** [backend/src/trinity/publication/service.py:131](#)
- **What it does:** Commits all publication effects together, or leaves every effect absent.
- **Why it is hard:** Idempotent finish under claim/generation; partial commit is forbidden. (CC=23)

35. RunPage

- **Score:** 5 · **CC:** 23 · **Nesting:** 1 · **LOC:** 33
- **Theme:** Publication / refresh
- **Where:** [frontend/src/pages/refresh/Refresh.tsx:122](#)
- **What it does:** Admin run detail: timeline, candidate panel, recovery actions, step paging.
- **Why it is hard:** Many concurrent queries/commands (recovery, steps cursor, cache invalidation). (CC=23)

36. _validate_reserved

- **Score:** 5 · **CC:** 22 · **Nesting:** 2 · **LOC:** 88
- **Theme:** EIA connector / validate
- **Where:** [backend/src/trinity/connector/validate.py:727](#)
- **What it does:** Runs and persists one reserved validation attempt; keeps failures without repair.
- **Why it is hard:** Long path that binds evidence prefixes, required checks, and diagnostics in one attempt. (CC=22, 88 lines)

37. EIAClient._collect

- **Score:** 5 · **CC:** 22 · **Nesting:** 3 · **LOC:** 68
- **Theme:** EIA connector / validate
- **Where:** [backend/src/trinity/connector/client.py:417](#)
- **What it does:** Pages EIA until empty page; accumulates unique rows or fails with no partial success.
- **Why it is hard:** Pagination, uniqueness, and fail-closed partials; timeout/page-limit interactions. (CC=22)

38. PublishedReader.read_into

- **Score:** 5 · **CC:** 22 · **Nesting:** 3 · **LOC:** 52
- **Theme:** SQL / query runtime
- **Where:** [backend/src/trinity/queries/staging.py:31](#)
- **What it does:** Reads published object bytes using only the trusted bundle encoding descriptor.
- **Why it is hard:** Compressed vs legacy paths; size/digest checks before callers see bytes. (CC=22)

39. check_preview

- **Score:** 5 · **CC:** 22 · **Nesting:** 4 · **LOC:** 38
- **Theme:** Auth / sessions
- **Where:** [backend/src/trinity/auth/preview_check.py:60](#)
- **What it does:** Verifies exact preview pages per role and Viewer detail denial.
- **Why it is hard:** Fixture must be real publication evidence; role matrix is strict. (CC=22)

40. CursorCodec.decode

- **Score:** 5 · **CC:** 22 · **Nesting:** 1 · **LOC:** 37
- **Theme:** SQL / query runtime
- **Where:** [backend/src/trinity/queries/cursors.py:142](#)
- **What it does:** Decodes signed preview/SQL cursors; rejects bad/extended tokens safely.
- **Why it is hard:** Four-part token parse plus HMAC; all failures share one safe code. (CC=22)

41. `timeline.<callback@92>`

- **Score:** 5 · **CC:** 21 · **Nesting:** 9 · **LOC:** 19
- **Theme:** Publication / refresh
- **Where:** [frontend/src/pages/refresh/Refresh.tsx:92](#)
- **What it does:** Maps refresh steps into per-stage timeline state and copy.
- **Why it is hard:** Nested stage/attempt rules including the awaiting_approval special case. (CC=21, nesting=9)

42. `QueryExecution.execute`

- **Score:** 5 · **CC:** 18 · **Nesting:** 3 · **LOC:** 147
- **Theme:** SQL / query runtime
- **Where:** [backend/src/trinity/queries/client.py:67](#)
- **What it does:** Stages published files, runs isolated query container, checks output envelope.
- **Why it is hard:** Long lifecycle (reserve → stage → run → cleanup); owned cleanup on every path. (CC=18, 147 lines)

43. `store_candidate`

- **Score:** 5 · **CC:** 18 · **Nesting:** 3 · **LOC:** 140
- **Theme:** EIA connector / validate
- **Where:** [backend/src/trinity/connector/pipeline.py:216](#)
- **What it does:** Uploads one locally validated candidate and returns a verified storage receipt.
- **Why it is hard:** Long upload/verify sequence with cancel and timeout paths. (CC=18, 140 lines)

44. `QueryResponse.exact_cells`

- **Score:** 5 · **CC:** 18 · **Nesting:** 8 · **LOC:** 16
- **Theme:** SQL / query runtime
- **Where:** [backend/src/trinity/queries/schemas.py:39](#)
- **What it does:** Validates SQL result cell types/nullability/row shape before return.
- **Why it is hard:** Compact but dense type matrix; truncation/row-count invariants. (CC=18, nesting=8)

Score 4 — High

1. `PublicationCommands.command`

- **Score:** 4 · **CC:** 21 · **Nesting:** 3 · **LOC:** 87
- **Theme:** Publication / refresh
- **Where:** [backend/src/trinity/publication/commands.py:23](#)
- **What it does:** Applies one candidate action (approve/recover path) or returns an original receipt.
- **Why it is hard:** Auth capability switches by action; fingerprint/ETag replay rules. (CC=21, 87 lines)

2. `check_schedule`

- **Score:** 4 · **CC:** 21 · **Nesting:** 2 · **LOC:** 54
- **Theme:** Auth / sessions
- **Where:** [backend/src/trinity/auth/check.py:61](#)
- **What it does:** Proves Admin can save schedule without starting a run; others get 403.
- **Why it is hard:** Multi-step ETag/history dance per role; easy to break ordering. (CC=21)

3. admin_context

- **Score:** 4 · **CC:** 21 · **Nesting:** 5 · **LOC:** 33
- **Theme:** Publication / refresh
- **Where:** [backend/src/trinity/refresh/service.py:156](#)
- **What it does:** Builds admin eligibility codes for UI without granting publish/recover permission.
- **Why it is hard:** Nested status/setup/warning cases; display-only but must stay conservative. (CC=21, nesting=5)

4. SettingsForm

- **Score:** 4 · **CC:** 20 · **Nesting:** 3 · **LOC:** 41
- **Theme:** Frontend UI
- **Where:** [frontend/src/pages/settings/Settings.tsx:27](#)
- **What it does:** Settings form for schedule/setup with ETag save and notices.
- **Why it is hard:** Setup vs edit modes, cache invalidation, and navigation side effects. (CC=20)

5. parse_preview_input

- **Score:** 4 · **CC:** 20 · **Nesting:** 5 · **LOC:** 29
- **Theme:** SQL / query runtime
- **Where:** [backend/src/trinity/queries/preview.py:107](#)
- **What it does:** Parses URL pairs into PreviewInput before rate debit; keeps duplicates.
- **Why it is hard:** No semantic cross-field checks here—easy to assume too much. (CC=20, nesting=5)

6. execute_preview

- **Score:** 4 · **CC:** 19 · **Nesting:** 3 · **LOC:** 57
- **Theme:** SQL / query runtime
- **Where:** [backend/src/trinity/queries/runtime/preview.py:17](#)
- **What it does:** Runs a closed preview over staged files; returns $\leq$ page_size rows plus lookahead.
- **Why it is hard:** Rechecks operation then registers files; typed literals including ID-like values. (CC=19)

7. request

- **Score:** 4 · **CC:** 19 · **Nesting:** 2 · **LOC:** 52
- **Theme:** Frontend UI
- **Where:** [frontend/src/api/client.ts:165](#)
- **What it does:** Frontend HTTP helper: headers, If-Match, credentials, typed ApiResult errors.
- **Why it is hard:** Many option branches (body, etag, parse failures) shared by all pages. (CC=19)

8. Contributions

- **Score:** 4 · **CC:** 19 · **Nesting:** 3 · **LOC:** 43
- **Theme:** Frontend UI
- **Where:** [frontend/src/pages/dashboard/Dashboard.tsx:162](#)
- **What it does:** Shows facility contributions for the selected national day with keep-previous behavior.
- **Why it is hard:** Loading/selection races while keeping prior contributions visible. (CC=19)

9. `validate_query`

- **Score:** 4 · **CC:** 19 · **Nesting:** 2 · **LOC:** 35
- **Theme:** SQL / query runtime
- **Where:** `backend/src/trinity/queries/runtime/sql_policy.py:372`
- **What it does:** Runs the full D01 SQL policy in a bounded subprocess and returns a normalized ValidatedQuery.
- **Why it is hard:** Orchestrates several policy steps; failures must stay safe and non-leaky. (CC=19)

10. `_binding`

- **Score:** 4 · **CC:** 19 · **Nesting:** 1 · **LOC:** 26
- **Theme:** Publication / refresh
- **Where:** `backend/src/trinity/refresh/registration.py:47`
- **What it does:** Compares measured candidate evidence to the worker's frozen expectations.
- **Why it is hard:** Manifest/policy/window fields must all match or handoff fails. (CC=19)

11. `FacilityChoices`

- **Score:** 4 · **CC:** 19 · **Nesting:** 1 · **LOC:** 24
- **Theme:** Frontend UI
- **Where:** `frontend/src/pages/catalog/Choices.tsx:26`
- **What it does:** Searchable facility picker bound to preview filters.
- **Why it is hard:** Debounced search, visit/cache keys, and busy disable paths. (CC=19)

12. `_owned`

- **Score:** 4 · **CC:** 19 · **Nesting:** 1 · **LOC:** 20
- **Theme:** Publication / refresh
- **Where:** `backend/src/trinity/refresh/registration.py:25`
- **What it does:** Locks control/run/version/step in order; rejects stale writers via fence.
- **Why it is hard:** Canonical lock order plus lease/fence checks; races are subtle. (CC=19)

13. `prepare_candidate`

- **Score:** 4 · **CC:** 18 · **Nesting:** 2 · **LOC:** 110
- **Theme:** EIA connector / validate
- **Where:** `backend/src/trinity/connector/pipeline.py:447`
- **What it does:** End-to-end: extract → freeze → validate → store for one new version.
- **Why it is hard:** Stage orchestration across network, disk, and S3; events and cancellation interleaved. (CC=18, 110 lines)

14. `EIAClient._recorded_request`

- **Score:** 4 · **CC:** 18 · **Nesting:** 2 · **LOC:** 77
- **Theme:** EIA connector / validate
- **Where:** `backend/src/trinity/connector/client.py:579`
- **What it does:** Sends one GET and optionally records sanitized attempt evidence.
- **Why it is hard:** Success and failure both must retain safe evidence without leaking credentials. (CC=18, 77 lines)

15. `persist_candidate`

- **Score:** 4 · **CC:** 18 · **Nesting:** 2 · **LOC:** 64
- **Theme:** Publication / refresh
- **Where:** [backend/src/trinity/refresh/registration.py:75](#)
- **What it does:** Writes verified candidate handoff on the caller's transaction (no commit here).
- **Why it is hard:** Exact replay rules; must not commit; binding checks precede writes. (CC=18)

16. `ChoiceService.prepare`

- **Score:** 4 · **CC:** 18 · **Nesting:** 2 · **LOC:** 52
- **Theme:** SQL / query runtime
- **Where:** [backend/src/trinity/queries/choice_service.py:32](#)
- **What it does:** Prepares a choices query: auth, pin, debit, then storage/runtime work.
- **Why it is hard:** Metadata transactions must close before storage; counted failures retained. (CC=18)

17. `build_preview_response`

- **Score:** 4 · **CC:** 18 · **Nesting:** 1 · **LOC:** 36
- **Theme:** SQL / query runtime
- **Where:** [backend/src/trinity/queries/preview_schemas.py:195](#)
- **What it does:** Builds a bounded wire preview page from $\leq \text{limit}+1$ encoded rows.
- **Why it is hard:** Cursor/has_more/diagnostics must stay consistent with trusted evidence. (CC=18)

18. `serialize_preview_rows`

- **Score:** 4 · **CC:** 18 · **Nesting:** 6 · **LOC:** 34
- **Theme:** SQL / query runtime
- **Where:** [backend/src/trinity/queries/preview_schemas.py:159](#)
- **What it does:** Encodes Arrow-like cell values to wire strings without float rounding.
- **Why it is hard:** Per-column exact encoding; silent conversion would corrupt traces. (CC=18, nesting=6)

19. `Dashboard`

- **Score:** 4 · **CC:** 18 · **Nesting:** 2 · **LOC:** 31
- **Theme:** Frontend UI
- **Where:** [frontend/src/pages/dashboard/Dashboard.tsx:52](#)
- **What it does:** Loads dashboard data for a date range and owns preset/custom range controls.
- **Why it is hard:** Query key and range state interactions; custom vs preset paths are easy to confuse. (CC=18)

20. `ChoiceOperation.__post_init__`

- **Score:** 4 · **CC:** 18 · **Nesting:** 3 · **LOC:** 17
- **Theme:** Other backend
- **Where:** [backend/src/trinity/contracts/choices.py:31](#)
- **What it does:** Validates choice operation protocol fields and reuses preview identity checks.
- **Why it is hard:** Delegates into PreviewOperation then adds choice-specific rejects. (CC=18)

21. GeneratorChoices

- **Score:** 4 · **CC:** 18 · **Nesting:** 1 · **LOC:** 12
- **Theme:** Frontend UI
- **Where:** [frontend/src/pages/catalog/Choices.tsx:50](#)
- **What it does:** Generator picker that depends on selected facility/filters.
- **Why it is hard:** Disabled/open state tied to facility presence and query keys. (CC=18)

22. check_persona

- **Score:** 4 · **CC:** 17 · **Nesting:** 2 · **LOC:** 35
- **Theme:** Auth / sessions
- **Where:** [backend/src/trinity/auth/check.py:117](#)
- **What it does:** Logs in a persona and optionally checks catalog/preview/schedule; logout on failure.
- **Why it is hard:** Branchy optional suites; cleanup logout must always be attempted.

23. read_context

- **Score:** 4 · **CC:** 17 · **Nesting:** 3 · **LOC:** 31
- **Theme:** Publication / refresh
- **Where:** [backend/src/trinity/refresh/repository.py:206](#)
- **What it does:** Reads occupied refresh lifecycle, candidate, and unresolved warning.
- **Why it is hard:** Several joined tables; absence vs empty warning meanings differ.

24. candidate_view

- **Score:** 4 · **CC:** 17 · **Nesting:** 5 · **LOC:** 20
- **Theme:** Publication / refresh
- **Where:** [backend/src/trinity/refresh/tracking.py:48](#)
- **What it does:** Derives candidate display state; never authorizes publication.
- **Why it is hard:** Eligible/approved/frozen flags combine; easy to confuse with real auth. (nesting=5)

25. execute_local

- **Score:** 4 · **CC:** 16 · **Nesting:** 4 · **LOC:** 48
- **Theme:** SQL / query runtime
- **Where:** [backend/src/trinity/queries/runtime/engine.py:63](#)
- **What it does:** Revalidates policy, registers one staged dataset, returns a bounded result fragment.
- **Why it is hard:** Supervisor limits are external; engine must preserve exact numbers and labels.

26. partial_results

- **Score:** 4 · **CC:** 16 · **Nesting:** 2 · **LOC:** 39
- **Theme:** Other backend
- **Where:** [backend/src/trinity/workers/refresh.py:31](#)
- **What it does:** Loads only complete journal lines with matching detail bytes after a kill.
- **Why it is hard:** Torn last line vs earlier complete evidence; must not invent success.

27. CandidatePanel

- **Score:** 4 · **CC:** 16 · **Nesting:** 1 · **LOC:** 34
- **Theme:** Publication / refresh
- **Where:** [frontend/src/pages/refresh/Refresh.tsx:156](#)
- **What it does:** Shows candidate review actions (approve/discard/retry) for a version.
- **Why it is hard:** Command and query revision coupling; disposition gates which buttons appear.

28. ValidatedQuery.from_bytes

- **Score:** 4 · **CC:** 16 · **Nesting:** 2 · **LOC:** 25
- **Theme:** Other backend
- **Where:** [backend/src/trinity/contracts/queries.py:35](#)
- **What it does:** Decodes internal ValidatedQuery bytes; rejects noncanonical/oversized/extended messages.
- **Why it is hard:** Strict key set and canonical JSON; security-sensitive parser.

29. actions

- **Score:** 4 · **CC:** 16 · **Nesting:** 1 · **LOC:** 24
- **Theme:** Publication / refresh
- **Where:** [backend/src/trinity/publication/checks.py:119](#)
- **What it does:** Returns approve / publication_retry / discard eligibility used by all command paths.
- **Why it is hard:** Conservative flags depend on run, version disposition, warnings, and prior events together.

30. _check_cells

- **Score:** 4 · **CC:** 16 · **Nesting:** 6 · **LOC:** 24
- **Theme:** SQL / query runtime
- **Where:** [backend/src/trinity/queries/preview_schemas.py:66](#)
- **What it does:** Checks preview row width, nullability, and cell types.
- **Why it is hard:** Nested per-cell validation; dense type rules. (nesting=6)

31. _parse

- **Score:** 4 · **CC:** 16 · **Nesting:** 3 · **LOC:** 22
- **Theme:** Other backend
- **Where:** [backend/src/trinity/dashboard/calculation.py:40](#)
- **What it does:** Parses dashboard query pairs into scalars after inspecting all occurrences.
- **Why it is hard:** Rejects dict/collapsed inputs; duplicate keys must not silently win.

32. custody

- **Score:** 4 · **CC:** 15 · **Nesting:** 1 · **LOC:** 37
- **Theme:** Publication / refresh
- **Where:** [backend/src/trinity/publication/custody.py:34](#)
- **What it does:** Exclusive flock on prepare root/lock inodes until SQL completes.
- **Why it is hard:** No symlink follow; missing files uncreatable; nonblocking flock races.

33. seed_personas

- **Score:** 4 · **CC:** 15 · **Nesting:** 3 · **LOC:** 30
- **Theme:** Auth / sessions
- **Where:** [backend/src/trinity/auth/seed.py:22](#)
- **What it does:** Creates missing local personas atomically; never overwrites existing ones.
- **Why it is hard:** Read-then-write under transaction; password source side effects.

34. `_freeze_reserved`

- **Score:** 4 · **CC:** 14 · **Nesting:** 3 · **LOC:** 73
- **Theme:** EIA connector / validate
- **Where:** [backend/src/trinity/connector/parquet.py:235](#)
- **What it does:** Freezes source evidence into a newly reserved directory; retains failures.
- **Why it is hard:** Multi-stage freeze (source → parquet); failure retention without repair. (73 lines)

35. `load_candidate`

- **Score:** 4 · **CC:** 14 · **Nesting:** 3 · **LOC:** 63
- **Theme:** Publication / refresh
- **Where:** [backend/src/trinity/refresh/evidence.py:54](#)
- **What it does:** Loads candidate only if parent completion, hashes, and local/remote bytes match.
- **Why it is hard:** Expected hash comes from durable custody, not a fresh hash of the file.

36. `EvidenceCache.read`

- **Score:** 4 · **CC:** 14 · **Nesting:** 3 · **LOC:** 57
- **Theme:** Publication / refresh
- **Where:** [backend/src/trinity/publication/evidence_cache.py:91](#)
- **What it does:** Joins or owns a bounded evidence fill; returns only the authorized dataset.
- **Why it is hard:** Cache fill races plus deadline checks; failures must not leave bad entries.

37. `StorageOperation.put_verified`

- **Score:** 4 · **CC:** 14 · **Nesting:** 4 · **LOC:** 54
- **Theme:** Storage / adapters
- **Where:** [backend/src/trinity/adapters/s3.py:343](#)
- **What it does:** Create-once put then read-back verify; rejects conflicts and unresolved writes.
- **Why it is hard:** Reservation tokens and conflict stops; ambiguous S3 outcomes are hard.

38. `rotate_passwords`

- **Score:** 4 · **CC:** 14 · **Nesting:** 2 · **LOC:** 47
- **Theme:** Auth / sessions
- **Where:** [backend/src/trinity/auth/rotate.py:49](#)
- **What it does:** Replaces password hashes for named personas in one transaction.
- **Why it is hard:** Per-name password_source errors vs DB update atomicity.

39. `StorageOperation.verify`

- **Score:** 4 · **CC:** 14 · **Nesting:** 3 · **LOC:** 43
- **Theme:** Storage / adapters
- **Where:** [backend/src/trinity/adapters/s3.py:299](#)
- **What it does:** Streamed GET hashed against size and SHA-256 (never ETag).
- **Why it is hard:** Missing-key vs bad-bytes vs retry exhaustion need different outcomes.

40. `validate_single_statement`

- **Score:** 4 · **CC:** 14 · **Nesting:** 1 · **LOC:** 39
- **Theme:** SQL / query runtime
- **Where:** [backend/src/trinity/queries/runtime/sql_policy.py:14](#)
- **What it does:** Parses input and requires exactly one nonempty SELECT root.
- **Why it is hard:** Semicolon/comment edge cases; parse errors must map to one safe validation failure.

41. RefreshList

- **Score:** 4 · **CC:** 14 · **Nesting:** 1 · **LOC:** 30
- **Theme:** Publication / refresh
- **Where:** [frontend/src/pages/refresh/Refresh.tsx:52](#)
- **What it does:** Paginated refresh run list with polling for admins.
- **Why it is hard:** Infinite query, refetch interval, and role gate.

42. ChoiceCursorCodec.decode

- **Score:** 4 · **CC:** 14 · **Nesting:** 1 · **LOC:** 29
- **Theme:** SQL / query runtime
- **Where:** [backend/src/trinity/queries/choices.py:100](#)
- **What it does:** Authenticates and decodes choice cursors; rejects other-purpose tokens.
- **Why it is hard:** Same token shape as preview but purpose-bound; mixups are security bugs.

43. PreviewResponse.consistent_page

- **Score:** 4 · **CC:** 14 · **Nesting:** 2 · **LOC:** 21
- **Theme:** SQL / query runtime
- **Where:** [backend/src/trinity/queries/preview_schemas.py:136](#)
- **What it does:** Post-condition: schema, counts, reason, and cursor agree for a page.
- **Why it is hard:** Many invalid combinations must raise; used as a trust boundary.

44. PreviewService.prepare

- **Score:** 4 · **CC:** 13 · **Nesting:** 2 · **LOC:** 61
- **Theme:** SQL / query runtime
- **Where:** [backend/src/trinity/queries/service.py:100](#)
- **What it does:** Debits rate after identity/shape checks, then pins publication and stages work.
- **Why it is hard:** Debit-even-on-later-failure rule; reauth before pin; transaction boundaries.

45. RecoveryService.command

- **Score:** 4 · **CC:** 13 · **Nesting:** 2 · **LOC:** 59
- **Theme:** Publication / refresh
- **Where:** [backend/src/trinity/refresh/recovery.py:109](#)
- **What it does:** Replays authorized recovery intent or atomically resolves the targeted failure.
- **Why it is hard:** Authority, syntax, receipt replay, and obsolete revisions all interact.

46. EIAClient._fetch_page

- **Score:** 4 · **CC:** 13 · **Nesting:** 1 · **LOC:** 54
- **Theme:** EIA connector / validate
- **Where:** [backend/src/trinity/connector/client.py:486](#)
- **What it does:** Fetches one bounded page, strips credentials, and checks source shape.
- **Why it is hard:** Arg checks before HTTP plus response validation; optional tracker adds branches.

47. cli

- **Score:** 4 · **CC:** 13 · **Nesting:** 1 · **LOC:** 46
- **Theme:** Publication / refresh
- **Where:** [backend/src/trinity/publication/recovery.py:104](#)
- **What it does:** CLI entry for publication recovery; no Redis/S3/EIA init; no raw exception prints.
- **Why it is hard:** Ops-facing argument/env parsing with strict safe output rules.

48. `verify_password`

- **Score:** 4 · **CC:** 13 · **Nesting:** 2 · **LOC:** 38
- **Theme:** Auth / sessions
- **Where:** `backend/src/trinity/auth/passwords.py:48`
- **What it does:** Verifies password in a bounded child process with concurrency slots.
- **Why it is hard:** Acquire/reap/deadline paths; must not leak slots on failure.

49. `PublicationService.claim`

- **Score:** 4 · **CC:** 13 · **Nesting:** 1 · **LOC:** 32
- **Theme:** Publication / refresh
- **Where:** `backend/src/trinity/publication/service.py:83`
- **What it does:** Commits the unique generation marker before verification runs.
- **Why it is hard:** Locking run/version/active together; claim must be unique and replay-safe.

50. `parse_choice_input`

- **Score:** 4 · **CC:** 13 · **Nesting:** 3 · **LOC:** 29
- **Theme:** SQL / query runtime
- **Where:** `backend/src/trinity/queries/choices.py:31`
- **What it does:** Parses choice query pairs; preserves duplicates; search is facilities-only.
- **Why it is hard:** Duplicate/unknown key rules before semantics.

51. `_json_value`

- **Score:** 4 · **CC:** 13 · **Nesting:** 2 · **LOC:** 22
- **Theme:** Other backend
- **Where:** `backend/src/trinity/contracts/manifest.py:30`
- **What it does:** Canonical JSON conversion without float conversion or decimal rounding.
- **Why it is hard:** Decimal/date/finite checks; silent rounding would corrupt evidence.

52. `parse_page`

- **Score:** 4 · **CC:** 13 · **Nesting:** 2 · **LOC:** 22
- **Theme:** Publication / refresh
- **Where:** `backend/src/trinity/refresh/schemas.py:173`
- **What it does:** Parses refresh list/step page params; rejects repeats/unknowns.
- **Why it is hard:** Two modes (runs vs steps) share logic with different param names.

53. `PreviewInput.__post_init__`

- **Score:** 4 · **CC:** 13 · **Nesting:** 2 · **LOC:** 15
- **Theme:** SQL / query runtime
- **Where:** `backend/src/trinity/queries/preview.py:70`
- **What it does:** Validates preview dates, facility/generator IDs, and limit bounds.
- **Why it is hard:** Packed typed checks; invalid ID vs date failures look similar.

54. `candidateCopy`

- **Score:** 4 · **CC:** 13 · **Nesting:** 1 · **LOC:** 12
- **Theme:** Frontend UI
- **Where:** `frontend/src/lib/copy.ts:70`
- **What it does:** Maps candidate disposition/review status to user-facing explanation text.
- **Why it is hard:** Many disposition/status combinations; copy must match publication rules.

55. RefreshCursorCodec._validate

- **Score:** 4 · **CC:** 13 · **Nesting:** 1 · **LOC:** 10
- **Theme:** Publication / refresh
- **Where:** [backend/src/trinity/refresh/cursors.py:63](#)
- **What it does:** Validates refresh cursor body counters against bigint and page size.
- **Why it is hard:** Strict key set plus purpose/target binding in a tiny function.

56. SessionProvider

- **Score:** 4 · **CC:** 1 · **Nesting:** 0 · **LOC:** 124
- **Theme:** Frontend UI
- **Where:** [frontend/src/session/SessionProvider.tsx:18](#)
- **What it does:** Holds login session, me payload, cache clear on sign-out, and auth refresh.
- **Why it is hard:** Long provider: loading/error/reason and cache coupling across the app. (124 lines)